



DP-603: Implement Real-Time Intelligence with Microsoft Fabric

Exercise 2 : Ingest real-time data with Eventstream in Microsoft Fabric

- **Create a workspace**
- **Create an eventhouse**
- **Create an Eventstream**
- **Add a source**
- **Add a destination**
- **Query captured data**





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Exercise 2 : Ingest real-time data with Eventstream in Microsoft Fabric

- **Transform event data**
- **Query the transformed data**



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Create a workspace

Create a workspace

Name *

Ingest RT data in MSF

This name is available

Description

Describe this workspace

Domain ⓘ

Assign to a domain (optional)

Learn more about workspace settings

Workspace image

Upload

Reset

Advanced ^

Contact list * ⓘ

d(dagstudentsgreatlearning (Owner) X

Workspace type ⓘ

What a user can do in a workspace depends on the individual user license(s) assigned to that user and the workspace type. [Learn more](#)

Power BI only workspace types

Power BI workspaces only support the use of Power BI capabilities.

Apply

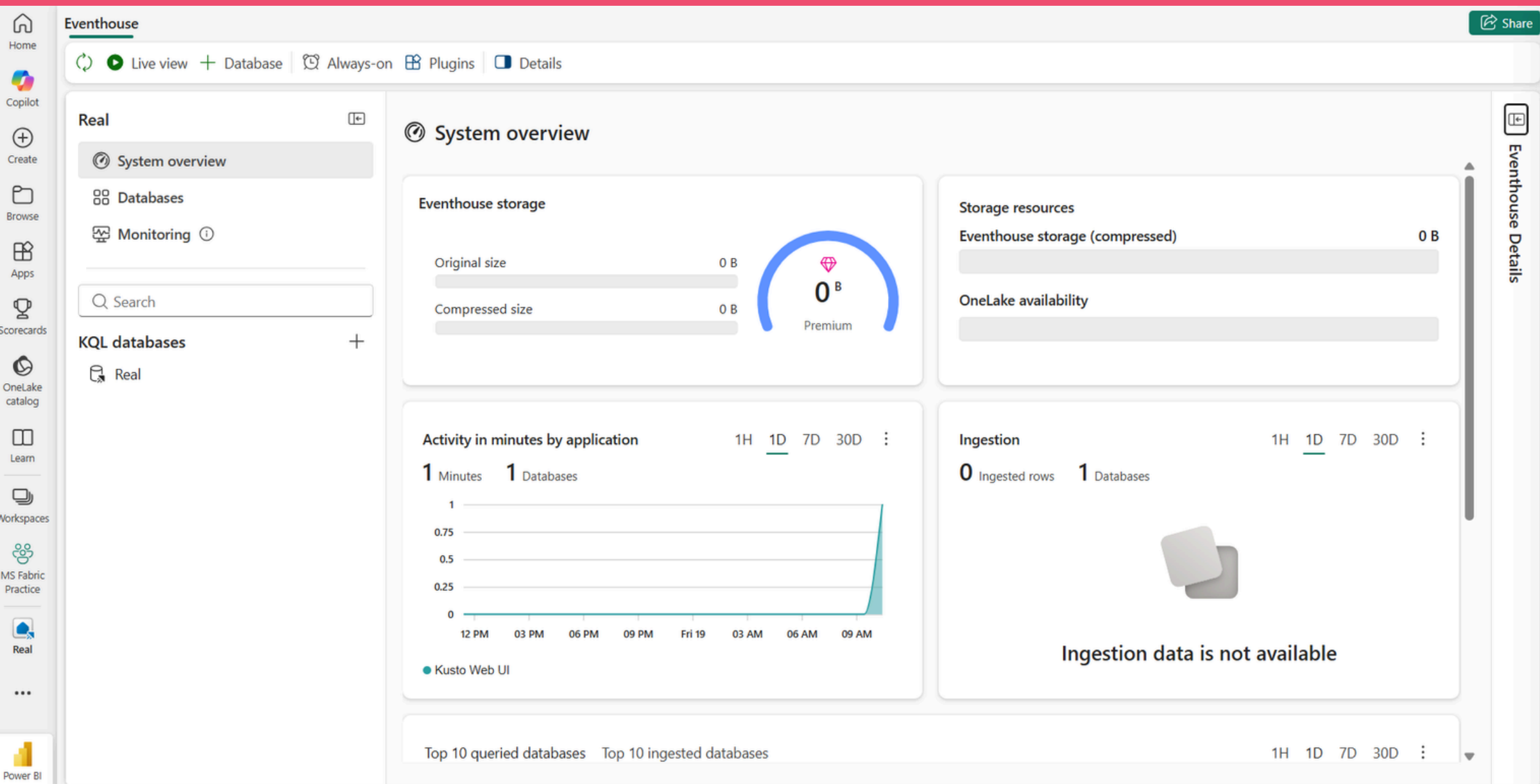
Cancel



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Create an eventhouse





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Create an Eventstream

Source

Configure

Inspect

Summary

Get data

Select a data source

Local file

Ingest up to 1000 files

Azure Storage

Azure Data Lake Storage Gen2/blob container or individual blobs

Amazon S3

S3 bucket or individual objects

Event Hubs

Ingest events in near real-time from an Azure Event Hub

Eventstream

Ingest events from an existing Eventstream

OneLake

Ingest a file from the OneLake catalog

Pipeline

Ingest data using Fabric Pipeline

Dataflow

Ingest data using Fabric Dataflow

Connect more data sources

Browse and connect data sources from Real-Time Hub

Real-Time hub

OneLake catalog

Filter by

Data type

Item owner

Item

Workspace

Search for streaming data

Data	Source item	Item owner	Workspace	Endorsement
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Cancel

New Eventstream



Enhanced Eventstream optimizes event usage by reusing events, routing based on content, and more. [Learn more](#)

Name *

Bicycle-data

Create

Cancel



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Add a source

The screenshot displays the Microsoft Fabric Data Engineering (DE) interface. The top navigation bar includes 'Home', 'Copilot', 'Create', 'Browse', 'Apps', 'Scorecards', 'Workspaces', 'MS Fabric Practice', and 'Bicycle-data'. The main workspace shows a data pipeline with three components: 'Bicycles' (source), 'Bicycle-data' (intermediate stream), and 'Transform events or add destination' (destination). The 'Bicycle-data' component is expanded, showing 'Bicycle-data-stream'. Below the pipeline, the 'Data preview' window is open, showing settings for 'Bicycle-data'. The 'Data type' is set to 'Json', 'Show data from' is set to 'Last hour', and the 'Time range' is from '06/19/26 10:42:55 AM' to '06/19/26 11:42:55 AM'. The preview area displays a message: 'Unable to preview data'.



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Add a destination

Home

Undo Redo Set alert Add source Transform events Add destination Setup required before publishing. Publish

Edit mode Changes will go live once you publish them. [Learn more](#)

Bicycles

Bicycle-data

Bicycle-data-stream

Eventhouse

This destination lets you ingest your real-time event data into an eventhouse, where you can use the powerful Kusto Query Language (KQL) to query and analyze the data.

bikes-table

Workspace *

MS Fabric Practice

Eventhouse *

bikes

Create new

KQL Database *

Type to select a KQL database

KQL Destination table *

Type to select a kusto table

Create new

Input data format * ⓘ

Json

☒ Activate ingestion after adding the data source

Save

Test result Authoring errors Refresh

Hide settings

Show data from

Last hour

Data preview not started

Data preview not started, please click refresh button to start data preview

Data

Sources

Bicycles

Streams

Bicycle-data-stream

Operators

No Operators

Destinations

bikes-table

Bicycles

Active

Bicycle-data

Bicycle-data-stream

bikes-table

Active

Details Data preview Refresh

Hide settings

Data type

Json

Last refreshed

06/19/26 11:49:36 AM

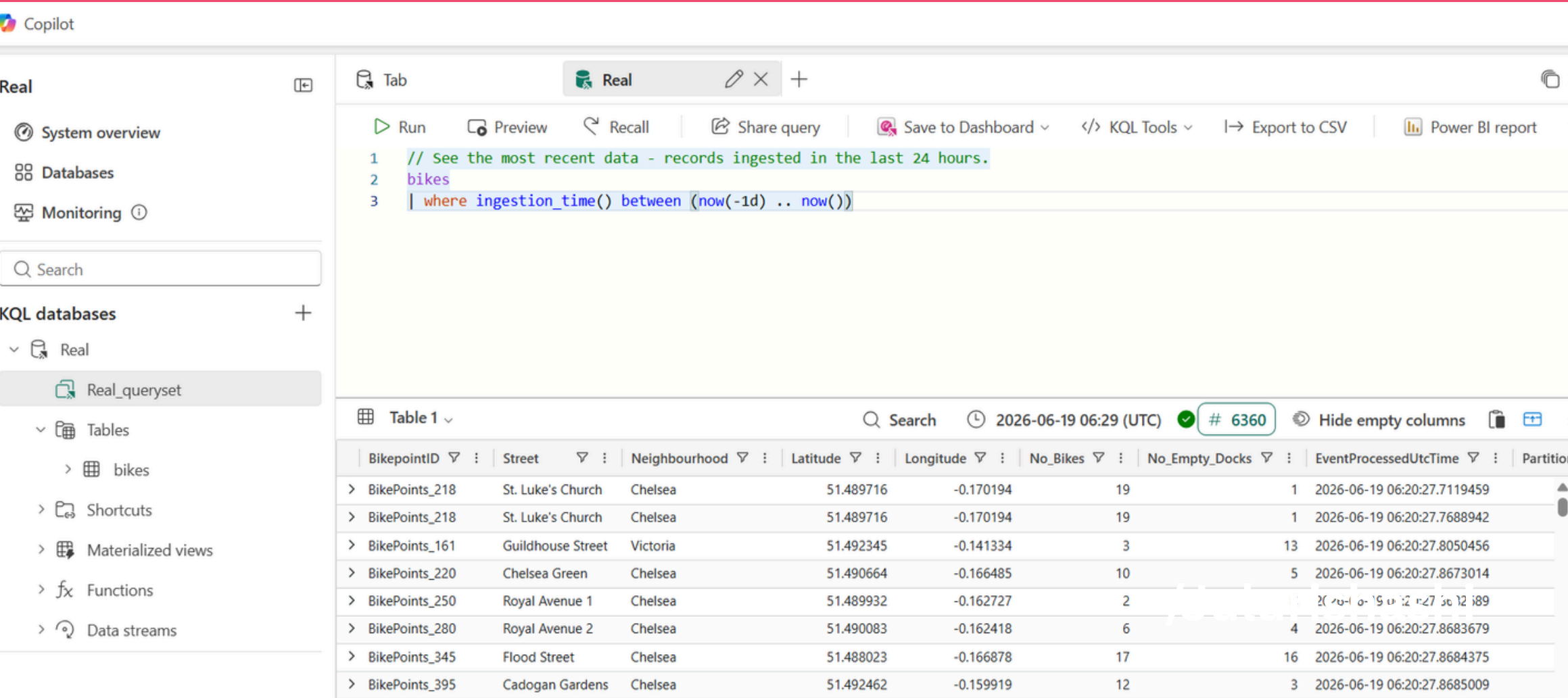
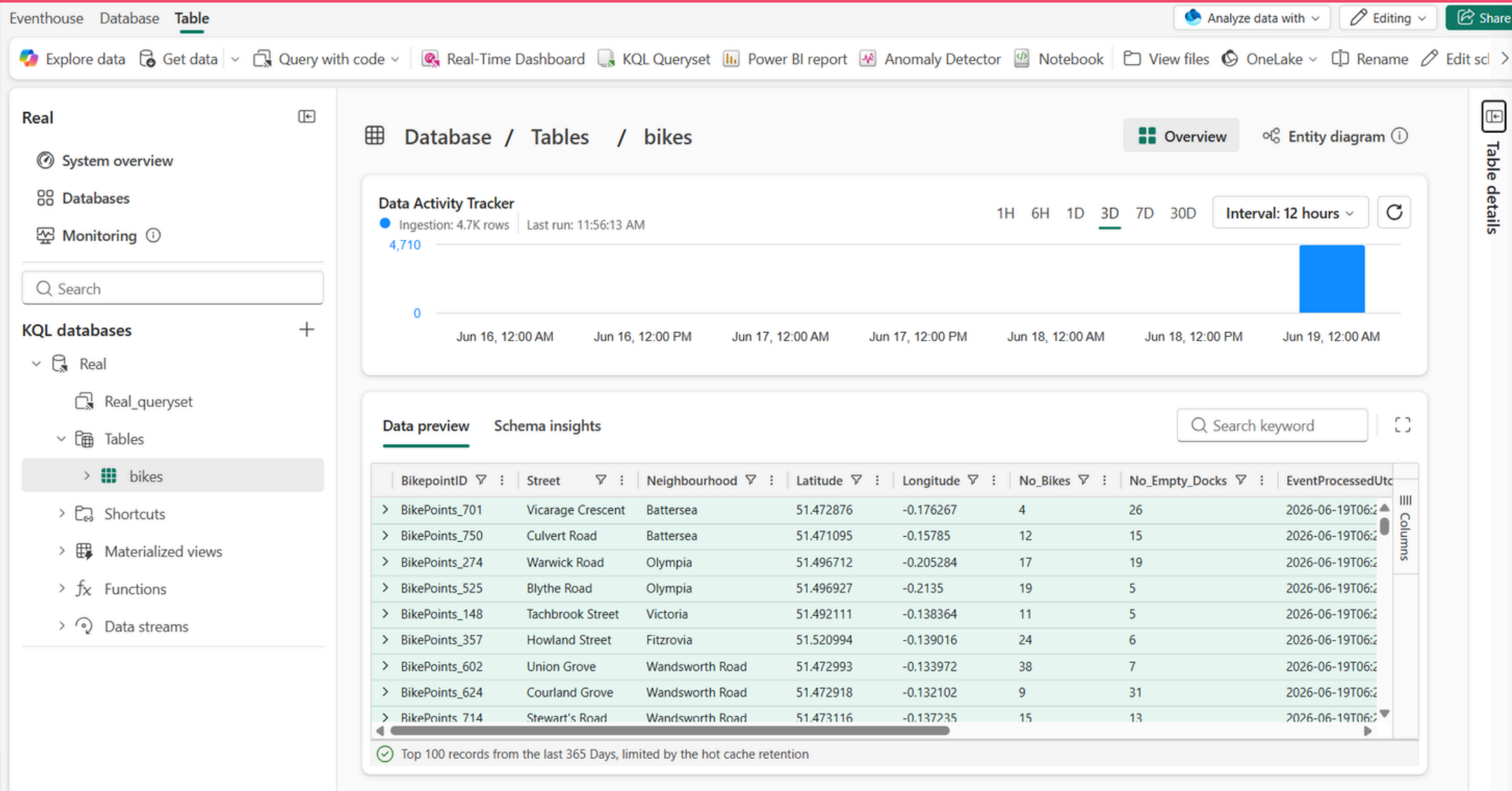
BikepointID	Street	Neigh...	Latitude	Longit...	No_Bik...	No_Em...
BikePoints_460	Burdett Road	Mile End	51.516196	-0.029138	28	5
BikePoints_714	Stewart's Road	Wandsworth Road	51.473116	-0.137235	15	13
BikePoints_624	Courland Grove	Wandsworth Road	51.472918	-0.132102	11	29
BikePoints_602	Union Grove	Wandsworth Road	51.472993	-0.133972	38	7
BikePoints_381	Charlotte Street	Fitzrovia	51.51953	-0.135777	8	6
BikePoints_357	Howland Street	Fitzrovia	51.520994	-0.139016	22	8
BikePoints_313	Wells Street	Fitzrovia	51.517344	-0.138072	22	16
BikePoints_311	Foley Street	Fitzrovia	51.519181	-0.140485	9	15
BikePoints_306	Rathbone Street	Fitzrovia	51.518162	-0.135025	13	3
BikePoints_81	Great Titchfield Str...	Fitzrovia	51.520253	-0.141327	7	12
BikePoints_28	Bolsover Street	Fitzrovia	51.523518	-0.143613	10	9



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Query captured data





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Transform event data

The screenshot shows a data engineering pipeline in Microsoft Fabric. The pipeline starts with a 'Bicycles' source, followed by a 'Bicycle-data' stream containing a 'Bicycle-data-stream' operator. The output of this stream is connected to a 'Group by' operator. The 'Group by' operator's settings are visible on the right:

- Operation name:** GroupByStreet
- Group aggregations by (optional):** Street
- Time window:** Tumbling
- Duration:** 5 Second
- Offset:** 0 Second

Below the pipeline, there is a 'No data to preview' message with the following text: "No data to preview. Possible reason: 1. Node is not configured. 2. No events fetched in source or upstream nodes, or result is empty. 3. There is no data at the time range 'Last hour'." A 'Refresh' button is located next to this message.

The screenshot shows a completed data engineering pipeline in Microsoft Fabric. The pipeline consists of the following components:

- Sources:** Bicycles
- Streams:** Bicycle-data-stream
- Operators:** GroupByStreet (configured with Tumbling window, Sum of No_Bikes aggregation)
- Destinations:** bikes-by-street-table, bikes-table

The pipeline is active, and the 'Data preview' tab is selected. The data preview shows the following table:

Street	SUM_No_Bikes	Window_End_Time
Maplin Street	13	2026-06-19T07:02:45Z
Kensington Gore	14	2026-06-19T07:02:45Z
Park Street	6	2026-06-19T07:02:45Z
Aberdeen Place	3	2026-06-19T07:02:45Z
Seville Street	22	2026-06-19T07:02:45Z
Wendon Street	2	2026-06-19T07:02:45Z



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Query the transformed data

Copilot

Real

System overview

Databases

Monitoring

Search

KQL databases

Real

Real_queryset

Tables

bikes

bikes-by-street

Ingestion Mappings

Update Policy

Street

SUM_No_Bikes

TabRealReal

RunPreviewRecallShare querySave to DashboardKQL ToolsExport to

1 // See the most recent data - records ingested in the last 24 hours.

2

3

4 ['bikes-by-street']

5 | take 100

6

Table 1

Search2026-06-19 07:10 (UTC)100

Street	SUM_No_Bikes	Window_End_Time
Poured Lines	5	2026-06-19 07:02:45.000
Mile End Stadium	13	2026-06-19 07:02:45.000
Grove End Road	7	2026-06-19 07:02:45.000
Lavington Street	18	2026-06-19 07:02:45.000
Wendon Street	2	2026-06-19 07:02:45.000
Seville Street	22	2026-06-19 07:02:45.000
Aberdeen Place	3	2026-06-19 07:02:45.000
Park Street	6	2026-06-19 07:02:45.000
Kensington Gore	14	2026-06-19 07:02:45.000
Maplin Street	13	2026-06-19 07:02:45.000
Stephendale Road	18	2026-06-19 07:02:45.000
Twig Folly Bridge	0	2026-06-19 07:02:45.000
Charlbert Street	8	2026-06-19 07:02:45.000
Denyer Street	21	2026-06-19 07:02:45.000
Charles II Street	18	2026-06-19 07:02:45.000



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Real

Run

Preview

Recall

Share query

Save to Dashboard

KQL Tools

Export to CSV

Power BI

```
1
2
3 ['bikes-by-street']
4 | summarize TotalBikes = sum(tolong(SUM_No_Bikes)) by Window_End_Time, Street
5 | sort by Window_End_Time desc, Street asc
```

Table 1

Search

2026-06-19 07:12 (UTC)

7440

Hide empty columns

Window_End_Time	Street	TotalBikes
> 2026-06-19 07:12:30.000	Allington Street	12
> 2026-06-19 07:12:30.000	Antill Road	5
> 2026-06-19 07:12:30.000	Ashley Place	16
> 2026-06-19 07:12:30.000	Belgrave Road	9
> 2026-06-19 07:12:30.000	Belgrave Square	4
> 2026-06-19 07:12:30.000	Blythe Road	17
> 2026-06-19 07:12:30.000	Bolsover Street	8
> 2026-06-19 07:12:30.000	Bourne Street	12
> 2026-06-19 07:12:30.000	Burdett Road	27
> 2026-06-19 07:12:30.000	Charlbert Street	7
> 2026-06-19 07:12:30.000	Charles II Street	21
> 2026-06-19 07:12:30.000	Charlotte Street	12
> 2026-06-19 07:12:30.000	Clinton Road	26
> 2026-06-19 07:12:30.000	Courland Grove	5
> 2026-06-19 07:12:30.000	Craven Street	6



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► Reference

